

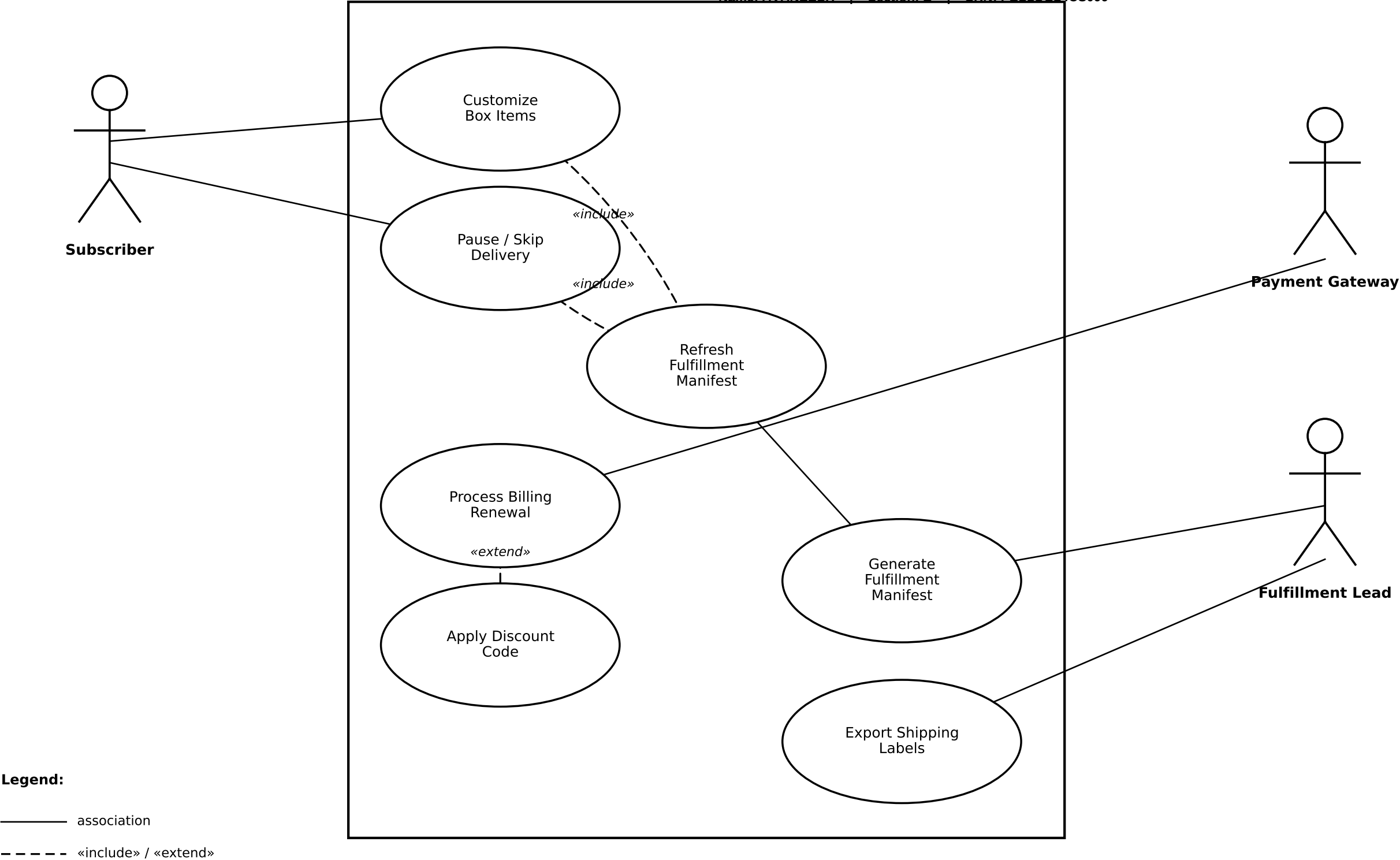
Requirements Table — Customizable Subscription Box Scheduler

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall allow subscribers to swap monthly box items or pause deliveries up to 48 hours prior to the monthly billing renewal date.	High	Pass: Selection updated and fulfillment manifest refreshed. Fail: Customization locked out prematurely.	Core flexibility feature required by the problem statement; protects subscriber control over recurring purchases.
FR-002	Functional	The system shall allow subscribers to select and update box preference tags (e.g., books, coffee, snacks) at signup and at any time before the customization lock-out window.	High	Pass: Selected tags are saved and reflected in the subscriber's profile within 2 seconds. Fail: Tags are not saved or not displayed after selection.	Personalization of box contents is the core value proposition of the subscription service.
FR-003	Functional	The system shall automatically regenerate the fulfillment manifest whenever a subscriber's box customization, pause, or skip status changes.	High	Pass: Manifest reflects the change within 5 minutes of the update. Fail: Manifest still shows stale data after 5 minutes.	Ensures the Fulfillment Lead always packs and ships the subscriber's latest choices (supports FR-001/FR-002).
FR-004	Functional	The system shall charge the subscriber's stored payment method via the payment gateway on the scheduled renewal date for every active (non-paused) subscription.	High	Pass: A gateway transaction ID is logged for each active subscription on its renewal date. Fail: No charge attempt is recorded, or a paused subscription is charged.	Automated, accurate billing is required for recurring revenue and to respect subscriber pause/skip choices.
FR-005	Functional	The system shall allow the Fulfillment Lead to export shipping labels and a packing manifest for all confirmed boxes ahead of the shipping cutoff.	Medium	Pass: The exported file lists every confirmed order with a valid shipping address. Fail: Any confirmed order is missing from the export.	Enables the fulfillment team to prepare and ship boxes on schedule.
NFR-001	Nonfunctional	The fulfillment manifest generator must export shipping labels for 10,000 monthly boxes in under 1 minute.	High	Pass: Benchmarking tests confirm the target latency and security standards under simulated peak load.	Large subscriber batches must be processed quickly to meet monthly shipping deadlines.
NFR-002	Nonfunctional	The subscriber-facing customization portal shall return page load and save responses in under 2 seconds for 95% of requests under normal load.	Medium	Pass: Load testing shows a 95th-percentile response time under 2 seconds. Fail: 95th-percentile response time exceeds 2 seconds.	Subscribers customize within a tight 48-hour pre-renewal window, so a responsive UI is essential to avoid missed edits.

Note: FR-001 and NFR-001 are the instructor-provided sample requirements; FR-002–FR-005 and NFR-002 were elicited from the problem scenario.

Customizable Subscription Box Scheduler

Name: AVANEESH | Section: B | SRN: PES1UG24CS099



Use-Case Flow Specification

Use Case: UC-03 — Process Billing Renewal

Primary Actor(s): System (Scheduler), Payment Gateway

Secondary Actor(s): Subscriber, Fulfillment Lead

Related Requirements: FR-001, FR-003, FR-004

Preconditions

- The subscriber has an active subscription with a valid payment method on file.
- The current date matches the subscriber's scheduled monthly billing renewal date.
- The subscriber has not paused or skipped the current cycle at least 48 hours prior to the renewal date (per FR-001).

Postconditions

- Success: The subscription is renewed, payment is captured, the fulfillment manifest includes the subscriber's box for the current cycle, and a confirmation is sent.
- Failure: The subscription is flagged for retry or suspended, the subscriber and Fulfillment Lead are notified, and the box is excluded from the current fulfillment manifest.

Name: AVANEESH |

Main Success Scenario

1. The system identifies all subscriptions whose renewal date is today and whose status is Active (not paused or skipped).
2. The system retrieves the subscriber's current box customization and any applicable discount code.
3. The system calculates the total charge amount, applying any valid discount.
4. The system sends a charge request to the Payment Gateway.
5. The Payment Gateway authorizes and processes the payment.
6. The system records the successful transaction and updates the subscription status to "Renewed."
7. The system refreshes the fulfillment manifest (UC-06) so the box is included in the current shipping cycle.
8. The system sends a confirmation notification and billing receipt to the subscriber.
9. The use case ends successfully.

Alternate Flow — 5a. Payment Declined

- 5a1. The Payment Gateway returns a declined or failed transaction status.
- 5a2. The system notifies the subscriber of the failed payment and requests updated payment details.
- 5a3. The system retries the charge once, 24 hours later, using the updated payment details.
- 5a4. If the retry also fails, the system suspends the subscription, excludes the box from the fulfillment manifest, and notifies both the subscriber and the Fulfillment Lead.